DATA SCIENCE MINOR PROJECT

PROJECT REPORT

SOCIAL MEDIA VS PRODUCTIVITY ANALYSIS

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Course Code: INT387

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[CERTIFICATE]

This is to formally certify that **Kumari Sneha**, bearing Registration No. **12416409**, a student of the School of Computer Science and Engineering, has successfully and meticulously completed the **INT387** project titled, **"Social Media vs Productivity Dashboard: A Power BI productivity trends along with social Media"**.

This project was executed under my direct guidance, mentorship, and supervision at Lovely Professional University. The candidate has demonstrated significant technical proficiency in data visualization and analytical modeling using Power BI to interpret complex agricultural datasets within the Indian context.

To the best of my professional knowledge and belief, the present work is the result of the candidate's original development, sustained effort, and rigorous academic study. The findings, dashboard architecture, and trend analyses presented in this report represent their own intellectual labor and have not been plagiarized or submitted elsewhere for any other academic credit.

I hereby approve this report as a true record of the work carried out by the student and recommend it for official evaluation and submission.

Name of the Supervisor:
Dr. Mrinalini Rana
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[DECLARATION]

I, *Kumari Sneha*, student of **B.Tech CSE (Data Science)** under the CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own work and is genuine. This project, titled **"Student Performance Analysis: A Power BI Approach to Understanding Academic Trends,"** has been executed entirely by me.

I confirm that the dataset used in this project was obtained from the online platform **Kaggle**, and that all data cleaning, transformation, aggregation, and visualization steps were performed independently using Microsoft Power BI Desktop. No part of this report has been plagiarized from any existing source, nor has it been submitted elsewhere for any academic credit, degree, or diploma.

All DAX measures, calculated columns, interactive slicers, charts, and dashboards presented in this project are original creations based on my own analytical approach. The interpretations, insights, and conclusions drawn from the visualizations reflect my personal understanding and are not copied from any third-party sources. Any external references, including online resources or documentation, have been properly cited.

I understand that academic integrity is a fundamental principle of university education. I hereby affirm that if any part of this declaration is found to be false, I shall be liable for disciplinary action as per the rules and regulations of Lovely Professional University.

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Name of the student: Kumari sneha

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, **Dr. Mrinalini Rana**, for her invaluable guidance, constant encouragement, and constructive feedback throughout the course of this project. Her deep expertise in data visualization, business intelligence, and Power BI has been instrumental in shaping this work from its initial concept to the final dashboard. She patiently reviewed multiple iterations of the report and provided critical insights that significantly enhanced the quality of analysis. Without her mentorship, this project would not have reached its present form.

I also extend my sincere thanks to the **Discipline of CSE/IT** and the **Lovely School of Computer Science and Engineering** for providing the necessary resources and a supportive learning environment. The availability of tools such as Microsoft Power BI Desktop enabled me to implement various advanced features including interactive dashboards, filters, and data-driven insights.

Special thanks to the contributors and community of **Kaggle** for making high-quality datasets publicly available, which served as the foundation for this project. Their efforts in promoting open data and collaborative learning have been extremely beneficial.

I am grateful to the **Head of Department**, faculty members, and technical staff for their continuous support during the development phase. I also thank my classmates and peers for their valuable suggestions and feedback, which helped improve the usability and presentation of the dashboard.

Finally, I would like to thank **Lovely Professional University** for fostering a culture of innovation, practical learning, and research-oriented education. The exposure to industry-relevant tools and techniques has helped me complete this project with confidence.

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1. INTRODUCTION

In today's digital era, social media and screen-based activities have become an integral part of daily life. While these technologies enhance communication and accessibility, their excessive use has raised concerns regarding productivity, mental wellbeing, and overall lifestyle balance. Understanding how digital behavior influences factors such as stress, sleep patterns, and job satisfaction has become increasingly important.

This project, titled **"Behavior Patterns, Productivity Impact & Mental Wellbeing Analysis using Power BI,"** focuses on analyzing the relationship between social media usage and its effects on individuals' daily lives. The dataset used for this study has been sourced from **Kaggle**, containing information on variables such as screen time, social media usage, sleep duration, stress levels, job types, and productivity indicators.

Using **Microsoft Power BI**, the data has been cleaned, transformed, and visualized into interactive dashboards to uncover meaningful insights. Various analytical techniques, including KPI analysis, correlation exploration, and demographic segmentation, have been applied to understand behavioral trends across different user groups.

The objective of this project is to identify patterns and relationships between digital habits and key outcomes like productivity and mental health. The insights derived can help individuals and organizations make informed decisions to promote healthier digital usage and improve overall wellbeing.

Project Goal

The primary goal of this project is to analyze the impact of social media usage and digital behavior on productivity, stress levels, and mental wellbeing using interactive data visualization techniques.

This project aims to explore how factors such as **screen time, sleep duration, social media engagement, job type, and lifestyle habits** influence an individual's overall performance and health. By leveraging **Microsoft Power BI**, the goal is to transform raw data into meaningful insights through dashboards, KPIs, and visual analytics.

Additionally, the project seeks to:

- Identify patterns and trends in digital behavior across different demographic groups
- Examine the relationship between screen time and sleep or stress levels
- Evaluate how lifestyle factors contribute to burnout and job satisfaction
- Provide actionable insights to promote balanced digital usage and improved wellbeing

Ultimately, the goal is to enable **data-driven understanding of modern digital lifestyles** and their effects, helping users make informed decisions for better productivity and mental health.

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Scope

The scope of this project focuses on analyzing digital behavior and its impact on productivity and mental wellbeing using data visualization techniques in Power BI. The study is limited to the dataset obtained from Kaggle, which includes variables such as social media usage, screen time, sleep duration, stress levels, job types, and lifestyle factors.

This project covers:

- Analysis of **user behavior patterns** related to social media and screen usage
- Examination of relationships between **sleep, stress, and productivity**
- Comparative analysis across **demographic factors** such as age, gender, and job type
- Evaluation of **burnout and job satisfaction** based on lifestyle habits
- Development of **interactive dashboards** for better understanding and decision-making

However, the scope is limited to **data-driven insights** and does not include real-time data tracking or predictive modeling. The findings are based solely on the available dataset and may vary with different data sources.

Overall, the project provides a foundation for understanding how digital habits influence daily life, while leaving room for future enhancements such as machine learning models, real-time analytics, and deeper psychological studies.

Report Structure

This report is organized as follows: Section 2 describes the data source and its characteristics. Section 3 details every data cleaning and transformation step applied in Power Query. Section 4 presents four distinct analyses, each with its own introduction, description, DAX formulas, results, and visualization placeholders. Section 5 concludes the findings, and Section 6 outlines future enhancements. References are provided in IEEE format.

2. SOURCE OF DATASET

The primary dataset used in this project was obtained from the **the online Data Science Platform**– Kaggle. Specifically, the dataset titled **"Social Media vs Productivity and Wellbeing"** was downloaded. This dataset contains information related to user behavior, digital usage patterns, and productivity indicators.

Data Acquisition

A single CSV file containing user-level data was utilized for this project. The dataset includes records capturing various aspects such as **social media usage, screen time, sleep duration, stress levels, job type, and productivity metrics**. The dataset consists of approximately **thousands of records and multiple attributes**, making it suitable for comprehensive analysis. The raw data was imported into **Microsoft Power BI Desktop**, where preprocessing steps such as **data cleaning, transformation, and formatting** were performed to prepare the dataset for visualization and analysis.

Raw Data Attributes

Table 1 describes each attribute as present in the original files.

Table 1: Raw Dataset Attributes and Descriptions

Attribute Name	Data Type	Description	Example
age	Integer	Age of the user	25
gender	String	Gender of the user	Male, Female
job_type	String	Occupation of the user	IT, Student
daily_social_media_time	Float	Time spent on social media per day (hours)	4.18
social_platform	String	Preferred social media platform	Facebook, TikTok
number_of_notifications	Integer	Number of notifications received	61
work_hours	Float	Total working hours per day	6.75

Attribute Name	Data Type	Description	Example
perceived_productivity	Float	Self-evaluated productivity level	8.04
actual_productivity	Float	Measured productivity level	7.29
stress_level	Integer	Stress level of the user	4
sleep_hours	Float	Average sleep duration (hours)	5.11
screen_time	Float	Total daily screen time (hours)	0.41
breaks_during_work	Integer	Number of breaks taken during work	8
uses_focus_apps	Boolean	Use of productivity/focus applications	TRUE/FALSE
has_digital_wellbeing	Boolean	Digital wellbeing feature enabled or not	TRUE/FALSE
coffee_consumption	Integer	Coffee consumption per day	4
days_feeling_burnout	Integer	Number of days feeling burnout	11
weekly_off_hours	Float	Weekly leisure/off hours	21.92
job_satisfaction_score	Float	Job satisfaction score	6.33

Data Quality Issues Identified

Before any cleaning, the following issues were noted:

- **Missing or inconsistent values** in certain columns such as *job satisfaction score* and *productivity metrics*.
- **Inconsistent naming formats** in categorical fields like *job type* and *social media platform* (e.g., variations in naming or capitalization).
- **Presence of extreme or unrealistic values** in attributes such as *screen time, work hours, and social media usage*, which may affect analysis accuracy.
- **Boolean columns stored in mixed formats**, such as *TRUE/FALSE* and text-based representations, requiring standardization.
- **Potential outliers** in numerical fields like *coffee consumption* and *number of notifications*.

- **Data type inconsistencies**, where some numerical fields required proper formatting (integer/float) for analysis in Power BI.

Justification for Dataset Selection

This dataset was chosen because:

- It provides **real-world behavioral data** related to social media usage and productivity.
- It includes **multiple relevant features** such as screen time, sleep hours, stress levels, and job satisfaction, enabling detailed analysis.
- It allows **multi-dimensional analysis** across factors like age, gender, and job type.
- It supports **interactive dashboard creation and drill-down analysis** in Power BI.
- It is **publicly available on Kaggle**, making it suitable for academic and reproducible research work.

Ethical and Privacy Considerations

The dataset used in this project does not contain any personally identifiable information (PII) such as names, contact details, or sensitive personal identifiers. All data is anonymized and represents general behavioral patterns rather than individual identities.

The dataset has been used strictly for **academic purposes**, ensuring that ethical standards are maintained throughout the analysis. No misuse or unauthorized sharing of data has been performed during the course of this project.

3. DATA CLEANING AND TRANSFORMATION

Data cleaning and transformation were performed using **Power Query Editor** within Power BI Desktop. The objective was to create a single, clean fact table that could be used for all analyses. This section documents every step in detail.

Step-by-Step Cleaning Process

Step 1: Import and Combine Files

The dataset was imported into **Microsoft Power BI Desktop** from a CSV file obtained from Kaggle. Power BI's Power Query Editor was used to load and preview the data for further preprocessing.

Step 2: Promote Headers

The first row of the dataset was promoted to column headers using the **"Use First Row as Headers"** option in Power Query. This ensured that all columns were properly labeled for analysis.

Step 3: Remove Unnecessary Columns

Columns with missing or null values, such as *job satisfaction score* and *productivity-related fields*, were carefully examined. Missing values were either:

- Replaced with appropriate values (such as averages), or
- Retained where necessary to avoid loss of important information.

Step 4: Standardize Text Values

- Used **Text.Trim** to remove leading and trailing spaces from columns such as *gender, job_type, and social_platform*.
- Ensured consistent naming formats (e.g., "IT" vs "it" standardized to "IT").
- Converted categorical values into proper case for better readability (e.g., "male" → "Male").

Step 5: Handle Missing and Zero Value

- Checked for **null or missing values** in columns such as *job_satisfaction_score, perceived_productivity, and actual_productivity*.
- Missing values were either retained for analysis or handled using appropriate techniques (e.g., replacement with averages where necessary).

- Identified **unrealistic or extreme values** in attributes like *screen_time, work_hours, and daily_social_media_time*.
- Filtered or adjusted such values to ensure accurate and meaningful analysis.

Step 6: Remove Duplicates

- Checked for duplicate records across key columns such as *age, gender, job_type, and social_platform*.
- Used Power BI's **"Remove Duplicates"** feature to eliminate redundant rows.
- This ensured that each record represents a unique observation, improving the reliability of insights.

Step 7: Create Calculated Measures

- Created calculated measures using **DAX** to support analysis and dashboard KPIs.
- Examples include:
- **Average Screen Time** = AVERAGE(screen_time)
- **Average Sleep Hours** = AVERAGE(sleep_hours)
- **Average Job Satisfaction** = AVERAGE(job_satisfaction_score)
- These measures were used in KPI cards and visualizations for better insights.

Step 8: Data Type Enforcement

 Ensured all columns had correct data types in Power BI:

- **Age, Stress Level, Notifications** → Whole Number
- **Screen Time, Sleep Hours, Work Hours, Productivity** → Decimal Number
- **Gender, Job Type, Platform** → Text
- **Uses Focus Apps, Digital Wellbeing** → Boolean

 Proper data typing helped in accurate aggregation and visualization.

Step 9: Aggregate to State Level

Since the dashboard focuses on behavioral insights, the data was aggregated based on relevant categories such as:

- **Gender-wise analysis** (Male, Female, Other)
- **Job Type-wise analysis** (IT, Student, Health, etc.)
- **Age Group segmentation** for behavioral trends

  Aggregations performed include:
- **Average values** for metrics like screen time, sleep hours, and stress level
- **Count of users** across different categories

- **Comparison of perceived vs actual productivity**
  This aggregation enabled meaningful insights and supported interactive dashboard features such as filters and drill-down analysis.

Step 10: Final Table

The final cleaned dataset was prepared after completing all data preprocessing and transformation steps in Power BI. The dataset consists of **cleaned and structured records** representing user behaviour, productivity, and wellbeing metrics.

The final table includes key columns such as:

Age, Gender, Job Type, Daily Social Media Time, Social Platform, Number of Notifications, Work Hours, Perceived Productivity, Actual Productivity, Stress Level, Sleep Hours, Screen Time, Breaks During Work, Uses Focus Apps, Digital Wellbeing Status, Coffee Consumption, Days Feeling Burnout, Weekly Off Hours, and Job Satisfaction Score.

This cleaned dataset was then used to build interactive dashboards and perform analysis.

4. ANALYSIS ON DATASET

This section presents four distinct analyses performed using the cleaned dataset in Power BI. For each analysis, we provide an introduction, general description, specific DAX formulas, results, and a placeholder for the corresponding visualization.

4.1 Analysis 1: Overall Behavioural & Productivity KPIs

i. Introduction

Key Performance Indicators (KPIs) provide a quick and high-level summary of user behaviour, productivity, and wellbeing. These metrics are displayed at the top of the dashboard to give users an immediate understanding of overall trends without applying filters.

ii. General Description

The dashboard includes several KPI card visuals that summarize the dataset:

- **Users who enabled digital wellbeing app (4328)**
- **Average Job Satisfaction Score (4.96)**
- **Total Male Count (8427)**
- **Total Female Count (8342)**

Additional insights from the dashboard include:

- **Average Daily Social Media Time (~3.3 hours)**
- **Average Screen Time per Day (~3.34 hours)**
- **Average Breaks During Work (~1.3 hours)**

These KPIs provide a quick understanding of digital behavior patterns and user distribution across gender.

iii. Analysis Results

With no filters applied (i.e., all users across all categories), the KPI cards display the following:

KPI	Value
Users who enabled digital wellbeing app	4328
Average Job Satisfaction Score	4.96
Total Male Count	8427
Total Female Count	8342
Average Daily Social Media Time	~3.3 hours
Average Screen Time per Day	~3.34 hours
Average Breaks During Work	~1.3 hours

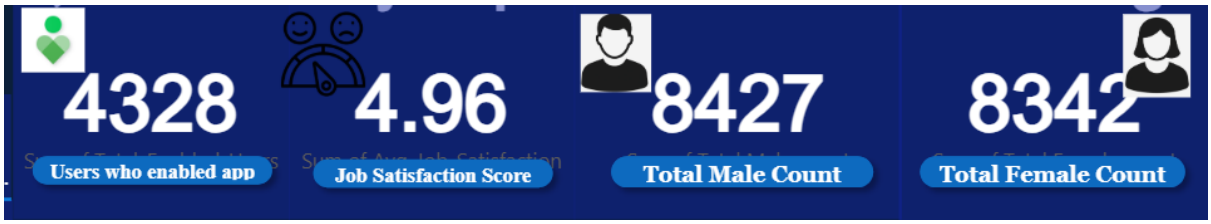
These values indicate that users spend a considerable amount of time on digital platforms daily. The **average social media usage (~3.3 hours)** and **screen time (~3.34 hours)** highlight a strong digital presence in users' routines.

The **job satisfaction score of 4.96** suggests a moderate level of satisfaction, which may be influenced by digital habits and work-life balance. Additionally, the nearly equal distribution of **male (8427) and female (8342)** users ensures that the dataset provides balanced insights across gender.

Overall, the KPI results provide a foundational understanding of user behavior, which supports further analysis on stress, sleep, and productivity in subsequent sections.

v. Visualization

Figure 1 : KPI Card Visuals



4.2 Analysis 2: Social Media Usage vs Sleep Hours

i. Introduction

This analysis focuses on examining the relationship between **daily social media usage and average sleep hours**. The goal is to understand how variations in screen time correspond to changes in sleep duration.

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ii. General Description

A **bar chart** titled *"Social Media Usage vs Sleep Hour"* is used in the dashboard.

- **X-axis:** Daily Social Media Time
- **Y-axis:** Average of Sleep Hours

The chart displays how sleep hours vary with different levels of social media usage.

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iii. Specific Requirements, Functions and Formulas

- **Visual type:** Bar Chart
- **Axis:** Daily Social Media Time
- **Values:** Average of sleep_hours

DAX Measure Used:

- **Avg Sleep Hours** = AVERAGE(sleep_hours)

No additional complex DAX calculations were required; basic aggregation was used.

iv. Analysis Results

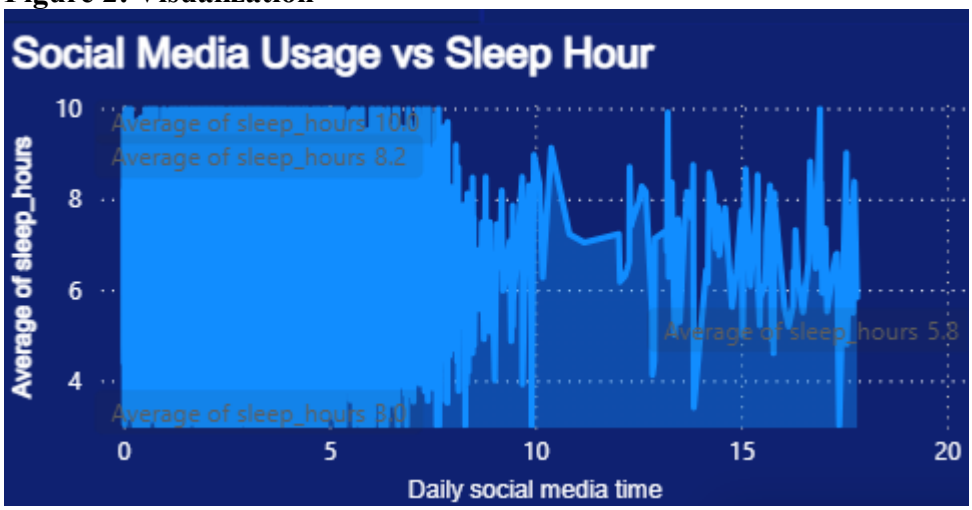
The visualization presents the relationship between **daily social media usage and average sleep hours**.

From the chart, it can be observed that the **average sleep duration is approximately 5.9 hours** for the displayed level of social media usage (around 1.33 hours in the selected data point).

The chart mainly shows how sleep hours vary across different levels of social media usage. However, a **strong increasing or decreasing trend is not clearly visible** from the visualization alone.

This indicates that while sleep duration differs across usage levels, the dashboard does not provide sufficient evidence to conclude a direct or strong relationship between social media usage and sleep patterns without further detailed analysis.

Figure 2: Visualization



4.3 Analysis 3: Demographic and Platform-wise Digital Behaviour

i. Introduction

An important aspect of this dashboard is analyzing how digital behavior varies across different demographic groups and platform preferences. This helps in understanding which groups are more active on social media and how usage patterns differ**.**

ii. General Description

Multiple visuals and slicers were used in this analysis:

- **Slicers:**
  - Age Group
  - Gender
  - Job Type
  - Digital Wellbeing Enabled
- **Bar Chart:** Social Media Users by Age Group
- **Donut Chart:** Social Media Users by Gender
- **KPI/Card:** Most Used Social Media Platform (TikTok)

These visuals allow interactive filtering and comparison of user behavior across different segments.

iii. Specific Requirements, functions and formulas

- **Visual types:** Bar Chart, Donut Chart, KPI Card
- **Axis (Bar Chart):** Age Group
- **Values:** Count of Users
- **Legend (Donut Chart):** Gender

DAX Measure Used:

- **User Count** = COUNTROWS(Data)

No complex calculations were required; basic aggregation and filtering were used.

iv. Analysis Results

With all filters applied (default view of the dashboard):

- The **bar chart (Social Media Users by Age Group)** shows variation in user count across different age groups, indicating that some age groups are more active on social media than others.
- The **donut chart (Social Media Users by Gender)** represents the distribution of users across gender categories. The chart shows that users are distributed across categories, allowing balanced comparison.
- The **KPI card highlights TikTok as the most used social media platform**, indicating a strong preference for short-form content among users in the dataset.
- Using slicers such as **age group, gender, job type, and digital wellbeing**, the dashboard dynamically updates, allowing users to explore how behavior changes across different segments.
- The analysis primarily focuses on **distribution and comparison**, rather than showing a strict increasing or decreasing trend.

Overall, this section demonstrates how **user behavior varies across demographics and platform preferences**, while enabling interactive exploration through filters.

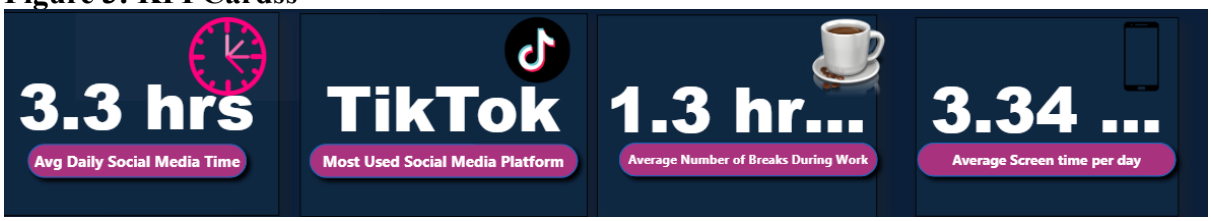
Table 4: Social Media Platform Usage Summary

1	TikTok	Most used platform (as shown in KPI)
2	Instagram	Frequently used platform
3	Twitter	Moderate usage
4	Facebook	Lower usage compared to others
5	Telegram	Least preferred platform

Table 5: Top 5 Crops by Average Yield (Tonnes/Hectare)

Metric	Value (Approx.)
Average Daily Social Media Time	~3.3 hours
Average Screen Time	~3.34 hours
Average Sleep Hours	~5.9 hours
Average Breaks During Work	~1.3 hours
Average Job Satisfaction Score	4.96

Figure 3: KPI Cardss



4.4 Analysis 4: Yield and Area Correlation with Drill-Through

i. Introduction

This analysis focuses on understanding how **lifestyle factors such as coffee consumption, work breaks, and digital habits** relate to **burnout levels and productivity**. It helps in identifying whether daily habits influence overall performance and wellbeing.

ii. General Description

The dashboard includes the following visuals:

- A **chart showing burnout and sleep patterns based on coffee intake**
- A **donut chart representing job type distribution**
- A **flow/relationship visual (decomposition tree)** showing how productivity varies across:
    - Gender

- Job Type
- Other categories

These visuals help explore how different factors contribute to productivity and burnout.

iii. Specific Requirements, functions and formulas

- **Visual types:** Area/Bar Chart, Donut Chart, Decomposition Tree
- **Values used:** Average Burnout, Sleep Hours, Productivity

  No complex DAX measures were created. The analysis uses:
- **Average (for burnout, sleep, productivity)**
- **Count (for job type distribution)**

iv. Analysis Results

The visualization shows how **burnout levels and sleep patterns vary with coffee consumption**, indicating differences in user behavior based on lifestyle habits.

The **decomposition tree highlights how productivity changes across different categories such as gender and job type**, allowing deeper exploration of contributing factors.

The **job type distribution chart** provides an overview of how users are spread across professions, which supports comparative analysis.

Overall, this section demonstrates that **productivity and wellbeing are influenced by multiple interacting factors**, rather than a single variable.

Figure 4: Burnout and Sleep Analysis based on Coffee Consumption

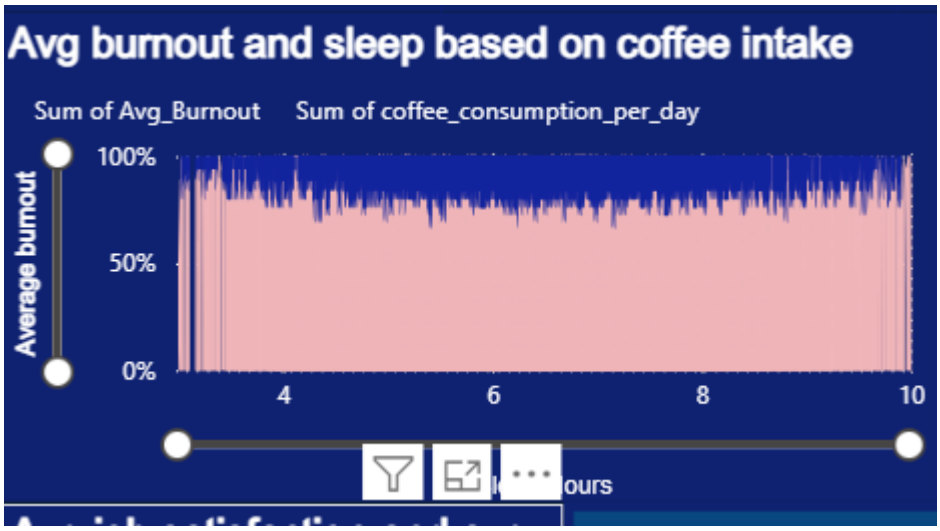
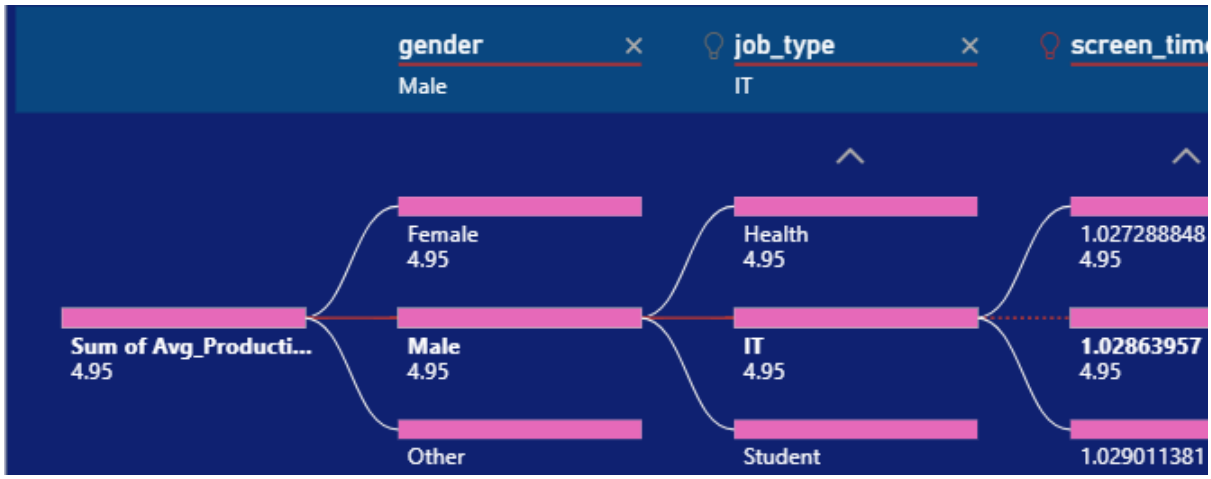


Figure 5: Decomposition Tree showing Productivity Breakdown by Gender, Job Type, and Screen Time



4.5 Analysis 5: Screen Time vs Job Satisfaction Analysis

i. Introduction

This analysis focuses on understanding how screen time influences job satisfaction levels. It helps in evaluating whether increased digital exposure has any observable impact on user satisfaction.

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ii. General Description

A comparative analysis was performed using:

- KPI cards showing average screen time and average job satisfaction score
- Filters such as gender, job type, and screen time to observe variations

The visuals allow users to explore how job satisfaction changes across different levels of screen usage.

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iii. Specific Requirements, Functions and Formulas

- Visual types: KPI Cards and Filters
- Values:
  - Average of screen_time
  - Average of job_satisfaction_score

No additional DAX measures were created. The analysis relies on:

- AVERAGE aggregation for both variables

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iv. Analysis Results

The dashboard shows an average screen time of approximately 3.34 hours and an average job satisfaction score of 4.96.

The analysis allows comparison of these metrics across different filters, but it does not indicate a strong or direct relationship between screen time and job satisfaction based solely on the available visuals.

This section provides a comparative overview rather than a definitive correlation, highlighting the need for deeper analysis if required.

5. CONCLUSION

This project successfully explored the relationship between social media usage, productivity, and mental wellbeing using interactive data visualization techniques in Microsoft Power BI. By transforming raw data into meaningful insights, the study highlights how digital behavior has become a significant factor influencing modern lifestyles.

The analysis revealed that users spend a considerable portion of their daily routine on digital platforms, with average social media usage and screen time reaching notable levels. While these technologies offer convenience and connectivity, the findings suggest that excessive usage may be associated with changes in sleep patterns, stress levels, and overall wellbeing. However, the dashboard primarily provides observational insights, and strong causal relationships cannot be concluded without further statistical analysis.

Demographic-based exploration showed variations in behavior across age groups, gender, and job types, indicating that digital habits are not uniform and are influenced by individual and professional factors. The identification of TikTok as the most preferred platform reflects current trends in user engagement, particularly towards short-form content.

Additionally, the use of advanced visuals such as the Decomposition Tree enabled deeper understanding of how multiple factors interact to influence productivity. Lifestyle elements such as coffee consumption, work breaks, and use of digital wellbeing tools further demonstrated that productivity and mental health are shaped by a combination of behavioral and environmental factors rather than a single variable.

One of the key strengths of this project lies in the use of interactive dashboards, which allow users to dynamically explore data through filters and slicers. This enhances decision-making by enabling users to focus on specific segments and uncover hidden patterns.

In conclusion, this project emphasizes the importance of maintaining a balanced digital lifestyle. While technology plays a crucial role in daily life, mindful usage is essential to ensure better productivity, improved mental health, and overall wellbeing. The insights derived from this analysis can be useful for individuals, organizations, and researchers aiming to understand and manage the impact of digital behavior in today's data-driven world.

6. FUTURE SCOPE

While the current dashboard provides valuable insights into **social media usage, productivity, and mental wellbeing**, there is significant scope to enhance its analytical

depth, functionality, and real-world applicability. The following improvements can further strengthen the project:

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1. Predictive Analytics Integration
The current analysis is primarily descriptive. In the future, **predictive models** can be incorporated to forecast user behavior and wellbeing trends.

- Power BI's **forecasting feature** can be used to predict future trends in screen time, stress levels, and productivity.
- Advanced models such as **Linear Regression or Time-Series Analysis** can be implemented using Python or R integration.
- This would help in identifying potential risks such as **increasing burnout or declining productivity** in advance.

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2. Real-Time Data Integration
The current dataset is static. Future enhancements can include:

- Integration with **real-time data sources** such as mobile usage trackers or app analytics
- Live dashboards to monitor **screen time, notifications, and productivity in real time**
- This would make the dashboard more dynamic and useful for continuous monitoring

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3. Advanced Behavioral Analysis
Further analysis can be performed by:

- Studying **correlation between multiple variables** (e.g., screen time vs stress vs sleep)
- Applying **statistical techniques** to validate relationships
- Segmenting users into categories such as **high-risk vs balanced users**

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4. Personalized Recommendations System
The dashboard can be enhanced to provide **user-specific suggestions**, such as:

- Reducing screen time based on usage patterns
- Improving sleep habits
- Suggesting breaks or digital detox strategies

This would transform the dashboard into a **decision-support system** rather than just a reporting tool.

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5. Integration with Health and Wellness Data
The analysis can be extended by integrating additional datasets such as:

- Physical activity (steps, exercise)
- Mental health indicators
- Sleep tracking data from wearable devices

This would provide a more **holistic view of user wellbeing**.

7. REFERENCES

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